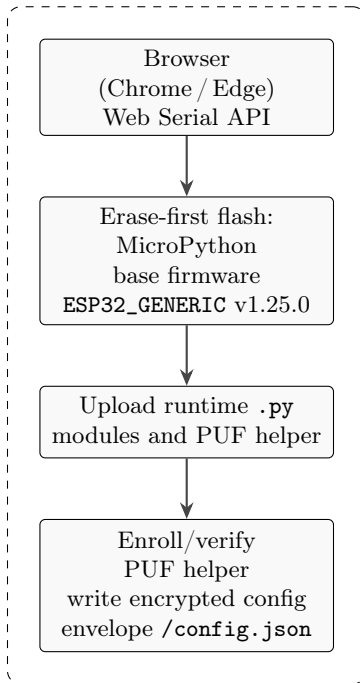


Provisioning (one-time)



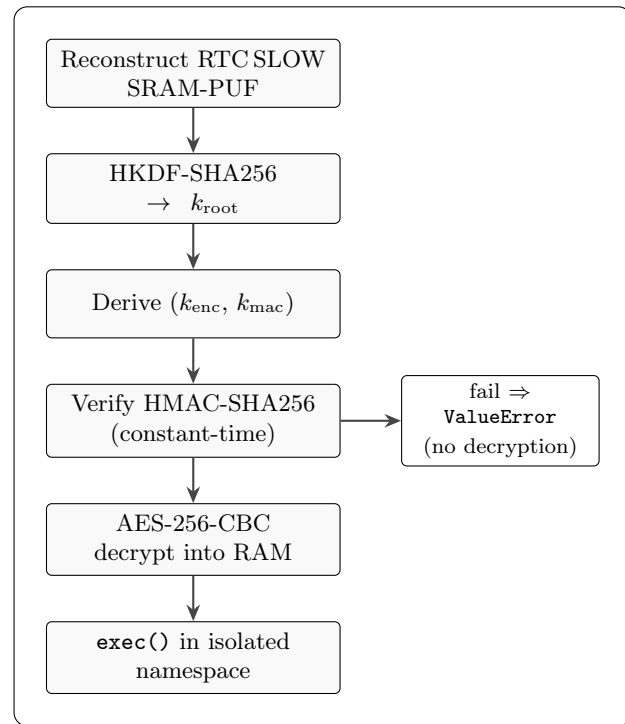
flash + modules + config envelope

ESP32-WROOM-32D (no
eFuse, no Secure Boot)
SRAM-PUF physical root

trusted key material

Host/browser: derives artifact
and config keys over raw REPL

Secure execution (per-invocation)



PUF root (runtime)